



Modeling and analysis of a methanol synthesis process using a mixed reforming reactor: Perspective on methanol production and CO₂ utilization



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HIGHLIGHTS

- A methanol synthesis process using a mixed reforming reactor was modeled.
- Kinetic models for the mixed reforming and methanol synthesis were developed.
- Various effects of operating conditions on the MeOH production rate were evaluated.
- An analysis was conducted with respect to both the overall and local CO₂ conversions.
- A trade-off existed between the maximum MeOH production and the maximum CO₂ utilization.

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ABSTRACT

In this study, kinetic models were developed for the mixed reforming and synthesis of methanol (MeOH). The effectiveness of the reforming model in our previous work was proven in an experimental study using a bench-scale reactor, while the intrinsic rate model and effectiveness factors were developed to represent the MeOH synthesis. For a 10-ton-per-day production of MeOH, the rate model was used to determine the size of a reforming reactor so that the supplied heat could be used exclusively to engage the reaction (not for the heat-up of the reactant), while the MeOH reactor was specified using the reported values. The process model was then used to evaluate various effects of the following factors on the MeOH production rate: (1) reaction temperature, (2) CO₂ fraction in the feed, and (3) the recycle route of the unreacted gas either to the feed or to the MeOH reactor. Additionally, an analysis was conducted with respect to both the overall and local CO₂ conversions in each reactor, and it was shown that a trade-off existed between the maximum MeOH production rate and the maximum CO₂ utilization, regardless of the existence of a recycle stream.

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1. Introduction

Methanol (MeOH) can be either used as a solvent and fuel by itself or conveniently converted into useful products such as formaldehyde, amines, acetic acid, esters, and olefins [1]. MeOH is considered as an excellent alternative energy resource since its high

octane number ensures good antiknock performance, in addition to high volatility, denser fuel–air charge and excellent lean burn properties [2]. In addition, it can be blended with gasoline, although it has half the volumetric energy density relative to gasoline or diesel [1]. Besides its direct use as fuel, methanol can be conveniently converted into ethylene or propylene in the MTO (methanol-to-olefins) process, and in turn, these olefins can be used to produce hydrocarbon fuels and their products [3]. Methanol is also used for supercritical treatment in the biodiesel fuel production [4–6], and is applied to the fuel cell system [7].

It is also a convenient medium for the storage and transport of CO and H₂ [8,9]. Recently, because of the increased interest in atmospheric pollution caused by the emission of significant

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amounts of greenhouse gases, particularly CO₂ [10], much attention has been focused on the development of processes for CO₂ utilization including MeOH synthesis by means of both CO and CO₂ hydrogenation [11–15]. In addition, if a MeOH production process is combined with the mixed (steam and CO₂) reforming of methane to produce syngas for MeOH synthesis, synergetic effects of CO₂ utilization are expected [16,17].

Much research work has been reported in the literature for the modeling and optimization of methanol synthesis processes. Holmgren et al. analyzed the energy balance of a commercial-scale MeOH process featuring a biomass gasification system [18]; in addition, the dynamic behavior and control of methanol synthesis fixed-bed reactors have been studied by many researchers [19,20]. Furthermore, the development of the methanol synthesis recycle-loop model has been described in detail, along with several case studies performed using steady-state and dynamic models for better understanding of the process behavior [21]. In addition to modeling and analysis, optimization methods have been extensively applied to the two processes including a genetic algorithm to obtain an optimal temperature profile and optimal two-stage cooling shell for the maximum production rate [22], and a repeated process estimation–optimization strategy has been applied to track the theoretical optimum profile of the selected control variable with the deactivation of the catalyst [23]. Optimal values of the inlet hydrogen mole fraction and the shell temperature have been investigated by employing methanol production rate as an objective function [24]; moreover, Luyben developed the economically optimum design of a methanol reactor and distillation column system to produce high-purity methanol from synthesis gas [25]. The syngas inlet temperature, steam drum pressure, and cooling water volumetric flow rate were optimized to maximize the methanol production in the reactor outlet [26]; Santangelo et al. presented an optimization procedure for increasing the methanol production in synthesis loops with quench reactors [27].

Recently, much effort has been directed to the reduction of CO₂ emissions to alleviate environmental phenomena such as global warming. In this sense, our previous work [17] reported the effects of operating conditions on the production rate of syngas and CO₂ conversion (utilization) for the mixed (steam + dry(CO₂)) reforming, where it was shown that a trade-off existed between the maximum syngas production and maximum CO₂ utilization. Based on these results, the mixed reforming reactor was incorporated into a MeOH synthesis process in the present work; here, the feed gas composed of CH₄, H₂O, and CO₂ was converted to syngas (CO + H₂ + CO₂) by a mixed reforming process, and the syngas was further converted to methanol by the hydrogenation of both CO and CO₂. This process includes both the consumption and production of CO₂, its consumption by dry reforming ($\text{CH}_4 + \text{CO}_2 \rightleftharpoons 2\text{CO} + 2\text{H}_2$), and CO₂ hydrogenation to synthesize MeOH ($\text{CO}_2 + 3\text{H}_2 \rightleftharpoons \text{CH}_3\text{OH} + \text{H}_2\text{O}$) vs. production by either steam reforming ($\text{CH}_4 + 2\text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + 4\text{H}_2$) or the water gas shift reaction ($\text{CO} + \text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + \text{H}_2$ (WGS)). Therefore, both an investigation of the effect of operating conditions on MeOH production and an analysis of the accompanying CO₂ emission were performed in this work.

2. Experimental

2.1. Bench-scale reforming reaction

The catalyst was prepared by a co-impregnation method using metal precursors of Ni(NO₃)₂·6H₂O (98%, Samchun) and Ce(CH₃COO)₃·xH₂O (99.9%, Aldrich) with an aqueous solution. The weight ratio of Ni/Ce/support (Sasol Pural MG30 with a weight ratio of MgO/Al₂O₃ = 3/7) was fixed at 15/6/79 based on the metallic composition of Ni and Ce. The prepared catalyst was dried

overnight at 393 K and pelletized in the form of a cuboid with four holes. Then, it was subsequently calcined at 1123 K for 6 h in air. Thereafter, the catalyst cuboid was crushed in a mortar, and then sieved to collect the final catalyst pellets of ca. 5 mm in size.

The prepared catalyst pellets were charged to a tubular reactor that had an outer diameter of 42.7 mm and an inner diameter of 32.5 mm. The amount of catalyst pellets was 40 g, and 1220 g of α-Al₂O₃ balls of ca. 5 mm in diameter was additionally mixed well with the catalyst pellets as a diluent material in the catalyst bed. After inert α-Al₂O₃ balls were charged to a height of 80 mm for pre-heating, the height of the catalyst bed was ca. 745 mm. The wall temperature was measured at three points, which were regularly positioned as shown in Fig. 1. Prior to the activity test, the catalyst was pre-reduced at 1023 K for 3 h under a flow of 5 vol% H₂ balanced with N₂. The gas-hourly-space-velocity (GHSV) was 25,000 mL-CH₄/(g_{cat} h), and the pressure was 0.53 MPa. The tube wall temperature was maintained at 1201 K by operating three jacket-type electric furnaces that surrounded the tube in a series. The feed molar ratio was specified as CH₄/H₂O/CO₂/N₂/H₂ = 4.07/6.37/1.75/1.00/1.07. The product gases were analyzed using an online gas chromatography unit (Younglin ACME 6100) equipped with a thermal conductivity detector connected to a Porapak-Q packed column for CO₂ and a molecular sieve 5A packed column for H₂, N₂, CO, and CH₄.

2.2. MeOH synthesis

A commercial catalyst (Cu/ZnO/Al₂O₃, Süd-Chemie, MegaMax 700) was used for the kinetic experiment on the synthesis of MeOH. The pelletized catalyst (5 mm diameter, 3 mm height) was broken and sieved to produce three samples of uniform size: 0.15–0.25 mm, 0.75–0.85 mm, and 1.5–2.5 mm that were labeled as S1, S2, and S3, respectively. Additionally, the original catalyst pellet, denoted as S4, was also used without breaking and sieving it. The S1 catalyst was used for the kinetic investigation, while the other catalyst samples were used to evaluate the effect of particle size on the catalytic performance.

The kinetic data were collected using a continuous tubular-flow fixed-bed microreactor. The temperature within the reactor was controlled by adjusting the furnace temperature, and the flow rate was controlled using a mass flow controller. The pressure was precisely controlled by a back pressure regulator and monitored by a digital pressure sensor. The catalyst samples (0.4 g) were diluted with similar-sized and inert α-Al₂O₃ particles (1.2 g) and packed together into a stainless steel reactor (internal diameter = 7 mm). The catalyst was reduced under a flow of H₂ (100 mL/min), and the temperature was increased from room temperature to a predetermined point at a rate of 2 K/min and held constant for 3 h. After the reduction, the pressure was increased according to the operating conditions, and the H₂ gas was then replaced by the gas used in the synthesis of MeOH. The detailed conditions of temperature, pressure, and feed gas composition are listed in Table S1 in Supplementary data. The reaction products were analyzed using an on-line gas chromatograph (Young Lin), where a Carboxen 1000 column was installed to separate the CO, CO₂, H₂, and Ar gases that were detected with the thermal conductivity detector, and a HP-PLOT Q capillary column coupled to a flame ionization detector was used to separate and analyze all the hydrocarbons, including MeOH.

3. Results and discussion

3.1. Reaction rates

The reaction rates for the mixed reforming process over the pellet-type catalysts in this work were reported in our previous work

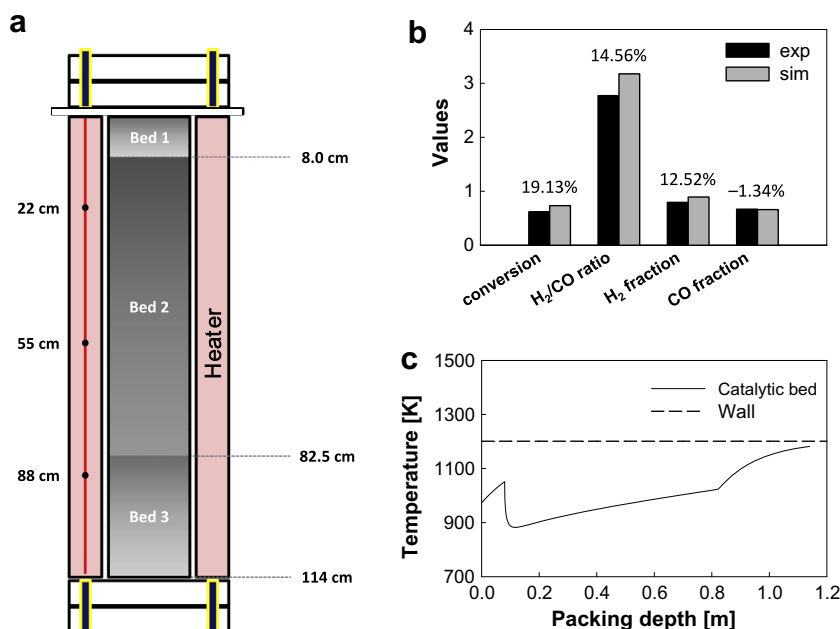


Fig. 1. (a) Scheme of a pilot-scale reforming reactor (tube I.D. = 3.25 cm, tube O.D. = 4.27 cm). Catalyst/inert weights are 0.00/0.14 kg, 0.04/1.22 kg (the ratio of inert to catalyst = 30.5), and 0.00/0.52 kg for Bed 1, 2, and 3, respectively, and the solid circles represent the location of the temperature detectors. (b) Comparison between experimental and simulation data: conversion = CH₄ conversion, H₂ fraction = H₂/(2CO + 3CO₂), and CO fraction = CO/(CO + CO₂) at the exit of the reactor. Relative residuals are shown above each bar, and the mean of absolute relative residuals (MARR) was determined to be 11.89%. (c) Temperature profile in the catalyst bed when the feed and wall temperatures are 973 and 1201 K, respectively.

[17]. To prove that the rates are representative of the reaction in large-scale reactors, the experiment was conducted using a tube larger than that used in the lab-scale experiments for kinetic study. Fig. 1a shows a schematic diagram of the reactor (detailed operating conditions are provided in Section 2.1 and the figure caption), and the simulation results are compared to experimental data in Fig. 1b. As shown in Fig. 1c, the temperature decreases abruptly at Bed 2 (0.08 m) although the fraction of catalyst is very low (inert/catalyst = 30.5), and remains below the wall temperature due to high reaction endothermicity. The CH₄ conversion at the exit was calculated to be 73.24% (experimental conversion = 61.48%), and the values of H₂/CO ratio, H₂ fraction (defined as H₂/(2CO + 3CO₂)), and CO fraction (defined as CO/(CO + CO₂)) were 3.18, 0.89, and 0.66, respectively (the corresponding experimental observations were 2.77, 0.79, and 0.67, respectively). The mean of the absolute relative residuals was 11.89%, indicating that the reaction rates in the previous work can be used for large scale reactors without alteration (see [17] for detailed reaction rates and kinetic parameters).

For the reaction rates of the MeOH synthesis, experiments were conducted using powder-type catalysts to generate the kinetic data. Experimental values for temperature, pressure, space velocity, CO fraction, and H₂ fraction were varied within the ranges 523–613 K, 5–9 MPa, 8000–40,000 mL/(g_{cat} h), 0–1, and 1.0–2.5, respectively, and the total number of experimental runs was 118 (see Table S1 in Supplementary data for detailed experimental conditions). To obtain the reaction rates for MeOH synthesis, the kinetic mechanism in the literature [11] was modified to represent the adsorption of CO and CO₂ on two different active sites (Cu¹⁺ and Cu⁰ for CO and CO₂, respectively) [12,13,28,29] while the assumption about the rate-determining step was maintained (see Table 1 for detailed reaction rates).

The kinetic parameters were estimated by fitting experimental data for 74 conditions (out of 118 total conditions) that deviated from the equilibrium conversions by more than 15% (the conditions marked with solid circles in Table S1 in Supplementary data),

and the estimation results are listed in Table 1. Since parameters estimation, especially in chemical kinetics, is very often related to strong multi-collinearity issues, and thus, the robustness of the selected procedure should be considered to prevent parameter estimates from significantly changing [30–32]. Therefore, in the present study, several initial values for the nonlinear regression including the reported values of kinetic parameters in the literature were considered to find the globally optimal estimates.

As shown in the comparison between experimental and simulation results (Fig. 2a), the model performs satisfactorily for CO conversion, while the mean of the absolute relative residuals (MARR) value for the CO₂ conversion was higher than that of the CO conversion. This result was attributed to the fact that the experimental values for the CO₂ conversion under the same conditions were close to zero, and thus the relative errors were calculated to be very large because of the very small denominator values. Additionally, when water and methanol in the reactor effluent were trapped to prevent them from entering the GC column, a small amount of CO₂ might have remained dissolved in the liquid phase, resulting in increased measurement errors. It was difficult to directly measure dissolved CO₂ in a blank test because a minimal amount of CO₂ was dissolved in the liquid phase compared with that in the gas phase (Nevertheless, there was a sufficient amount of CO₂ to affect the measurement errors of the CO₂ conversions). Thus, instead, we calculated the weight fractions of CO₂ in the liquid phase using a process simulator to determine the amount of CO₂ that could be dissolved in the liquid phase because of condensation. The values ranged from 0.14 wt% to 6.7 wt% under all operating conditions, indicating that the amount of dissolved CO₂ varied significantly with respect to the experimental conditions, which resulted in high individual errors in some cases.

The catalysts should be transformed into pellets to prevent an abrupt drop in pressure when using large-scale reactors. In such cases, the catalytic performance would deteriorate because of internal diffusion limitation, and accordingly, additional experiments were conducted using different mean particle sizes to

Table 1Reaction rates of MeOH synthesis and kinetic parameters estimated in this work.^a

Reaction	Rate equations ^b	Units
CO hydrogenation	$r_{CO} = \frac{k_A K_{CO} [f_{CO} f_{H_2}^{1.5} - f_{CH_3OH} / (K_{P,A} f_{H_2}^{0.5})]}{(1 + K_{CO} f_{CO}) (1 + K_{H_2}^{0.5} f_{H_2}^{0.5} + K_{H_2O} f_{H_2O})}$	mol/(kg _{cat} h)
Water gas shift reaction	$r_{WGS} = -\frac{k_B K_{CO_2} [f_{CO_2} f_{H_2} - f_{CO} f_{H_2O} / K_{P,B}]}{(1 + K_{CO_2} f_{CO_2}) (1 + K_{H_2}^{0.5} f_{H_2}^{0.5} + K_{H_2O} f_{H_2O})}$	
CO ₂ hydrogenation	$r_{CO_2} = \frac{k'_C K_{CO_2} [f_{CO_2} f_{H_2}^{1.5} - f_{H_2O} f_{CH_3OH} / (K_{P,C} f_{H_2}^{1.5})]}{(1 + K_{CO_2} f_{CO_2}) (1 + K_{H_2}^{0.5} f_{H_2}^{0.5} + K_{H_2O} f_{H_2O})}$	
DME production	$r_{DME} = \frac{k_{DME} K_{CH_3OH}^2 (C_{CH_3OH}^2 - (C_{H_2O} C_{DME}) / K_{P,DME})}{(1 + 2 \sqrt{K_{CH_3OH} C_{CH_3OH} + K_{H_2O} C_{DME}})^4}$	
Kinetic parameters	Estimated	Units
k'_A	$2.14 \times 10^4 \exp\left(\frac{-114,000}{RT}\right)$	mol/(kg _{cat} h Pa ^{1.5})
k'_B	$4.18 \times 10^8 \exp\left(\frac{-127,000}{RT}\right)$	mol/(kg _{cat} h Pa)
k'_C	$8.06 \exp\left(\frac{-68,000}{RT}\right)$	mol/(kg _{cat} h Pa ^{1.5})
k_{DME}	$3.07 \times 10^{13} \exp\left(\frac{-124,000}{RT}\right)$	mol/(kg _{cat} h)
K_{H_2O}	$3.80 \times 10^{-15} \exp\left(\frac{81,000}{RT}\right)$	Pa ⁻¹

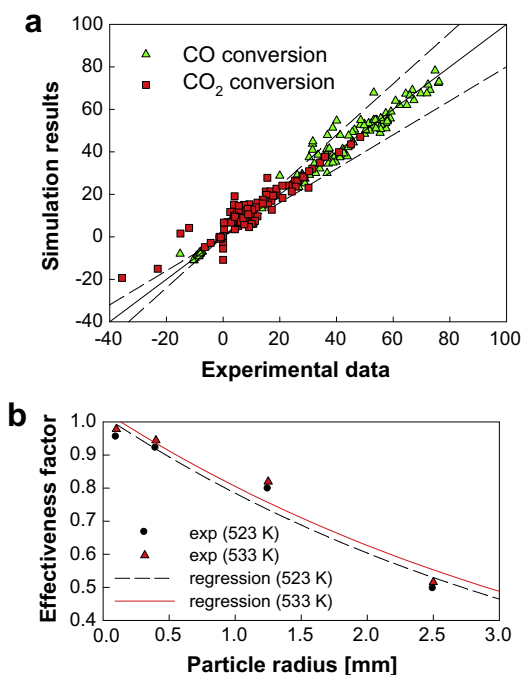
^a Other kinetic parameters were available in the literature and used without alteration [11,28,36].^b Units of fugacity (f_i) and concentration (C_i) are Pa and mol/m³, respectively. k , K_i , and $K_{P,j}$ are the reaction rate constant, adsorption equilibrium constant of species i , and equilibrium constant of reaction j , respectively. R is the gas constant (8.314 J/(mol K)).

Fig. 2. (a) Parity plots of CO (triangle) and CO₂ (square) conversion between experimental and simulated data. The values of mean of absolute relative residuals (MARR) for CO and CO₂ conversions were 9.40% and 72.12%, respectively. (b) The effectiveness factor was calculated using experimental data and regressed lines; $y = 1.019 \exp(-262.2x)$ for 523 K ($R^2 = 0.9552$) and $y = 1.034 \exp(-250.4x)$ for 533 K ($R^2 = 0.9692$).

evaluate the effects of mean particle size on the production rates. Based on the experimental data using catalysts in the group of S2–S4 (see the Experimental section for detailed particle sizes and Table S2 in Supplementary data for experimental values of CO conversion), the effectiveness factors for each particle size were calculated [33] and regressed using an exponent under the assumption that the factors approach zero as particle size increases. As shown in Fig. 2b, the effectiveness factors were slightly influenced by temperature, but substantially decreased as the particle radius increased, and they decreased to half when

the particle radius increased 25-fold (0.1–2.5 mm). Therefore, in a large-scale MeOH reactor, the value of the effectiveness factor was specified as 0.5.

3.2. Reactor sizing

A schematic of the MeOH synthesis process combined with the mixed reformer is provided in Fig. 3a, along with chemical species and overall reactions occurring in each reactor. In the industrial data [26], the reactor tube diameter and the length for MeOH synthesis reactor were specified as 0.04 m and 7 m, respectively, and the MeOH production rate from 1620 tubes was 11,283 kg-MeOH/h (270 ton-per-day (TPD)). Therefore, the same size was assumed for a tube while the number of tubes was reduced to 60 at a production rate of 10 TPD in this work. When the operating conditions were assumed to be a feed composition of CO/CO₂/H₂ = 13.9/13.1/73.0 (CO fraction $\approx$ 0.5, H₂ fraction $\approx$ 1.1), a pressure of 5 MPa, a reactor inlet temperature of 520 K, and a wall temperature of 483 K, the simulation of the MeOH reactor showed that $\sim$ 136 kmol/h of inlet flow comprising CO, CO₂, and H₂ is required to achieve an MeOH production rate of 10 TPD. Therefore, these conditions were specified as the reaction requirements. Notably, the CO fraction was assumed as 0.5 for simulating similar contributions from CO and CO₂ hydrogenation to the overall amount of MeOH synthesized. Since the temperature hot spot is a well-known key parameter in the MeOH synthesis, temperature profile in the reactor (graph not shown) was checked and the increase of ca. 10 K was observed at the location of approximately 1 m from the reactor inlet.

It is worth noting that, the kinetic model in the present study was based on the fresh (not aged) catalyst and the effectiveness factor was considered for the effects of particle size, since the operation duration of our pilot plant is limited. In the preliminary data, the deactivation of reforming catalyst was not observed, while detailed information about the deactivation of MeOH synthesis catalyst is not yet available. Therefore, the approach in the present study was applied to the process with fresh catalyst, and the deactivation was considered beyond the scope of the present study. However, further research is ongoing for the deactivation of MeOH synthesis catalyst through a long-term test, and the current approach will be applied to the design of a larger-scale

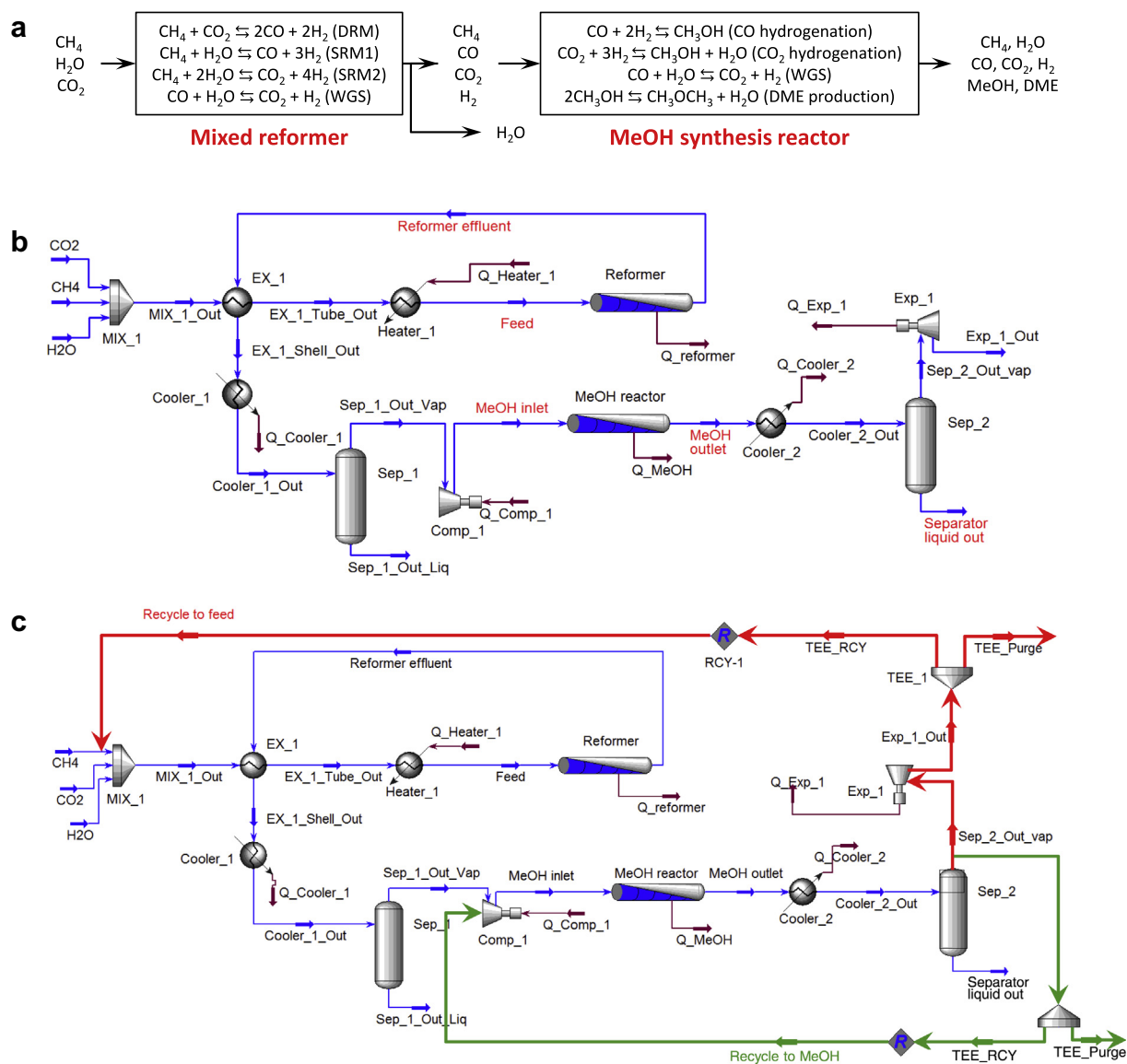


Fig. 3. (a) Schematic of the MeOH synthesis process combined with the mixed reformer and the process flow diagram of the MeOH synthesis plant (b) without and (c) with recycle in this work (recycle to the feed: red colored line, recycle to the MeOH inlet: green colored line). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

(demo- or commercial-scale process) process with the deactivation included for longer operation time.

It is also notable that, the radial temperature gradients were assumed to be negligible on the basis of our recent work [34], where small temperature differences in the radial direction were observed for the tubular reactor with similar reactor specification in the present study with the Fischer–Tropsch synthesis (FTS) reaction (its heat generation per unit volume was higher than MeOH reactor, due to higher exothermicity); Heat generation per unit volume in the FTS reactor [34] was $4.03 \times 10^5 \text{ W/m}^3$ at 540 K with the ratio of tube length (L) to diameter ($I.D.$) = 100 ($L = 5 \text{ m}$, $I.D. = 0.05 \text{ m}$) and overall heat transfer coefficient = $148 \text{ W/(m}^2 \text{ K)}$, while $1.08 \times 10^4 \text{ W/m}^3$ was generated at 480 K for the MeOH reactor in the present study (L/D ratio = 175 ($L = 7 \text{ m}$, $I.D. = 0.04 \text{ m}$), overall heat transfer coefficient = $140 \text{ W/(m}^2 \text{ K)}$).

In addition, the pre-heating temperature was specified to the feed temperature (1123 K) although metal dusting would prevent using any metallic heat-exchanger for the temperature. There are attempts to build specially designed feed-effluent heat-exchanger

for the reformer with catalytically active surfaces or sulphur (which is removed before the next reactor) included.

The equilibrium of the mixed reforming reaction was calculated for various feed compositions and reaction temperatures. This was done to help determine an appropriate size of the reformer and the optimal operating conditions for producing a gas mixture of ($\text{CO} + \text{CO}_2 + \text{H}_2$) with CO fraction and H_2 fraction close to 0.5 and 1.1, respectively, at a rate of 136 kmol/h. The calculation was conducted using the “Equilibrium Reactor” unit module in a process simulator (UniSim Design Suite, Honeywell Inc.), and it was found that the feed flow rate of 125 kmol/h with a composition of $\text{CH}_4/\text{CO}_2/\text{H}_2\text{O} = 28/6/66$, feed temperature = 1123 K, and a heat supply of 7 GJ/h resulted in the production of ($\text{CO} + \text{CO}_2 + \text{H}_2$) at a rate of 135.6 kmol/h with a CO fraction of 0.7 and a H_2 fraction of 1.12. Then, the detailed size of a tube for the reformer was determined so that the supplied heat was used exclusively by the reaction and not to heat the reactor effluent. Because the reforming reaction easily achieves equilibrium, even at substantially large space velocities, the contribution of axial convection was assumed to

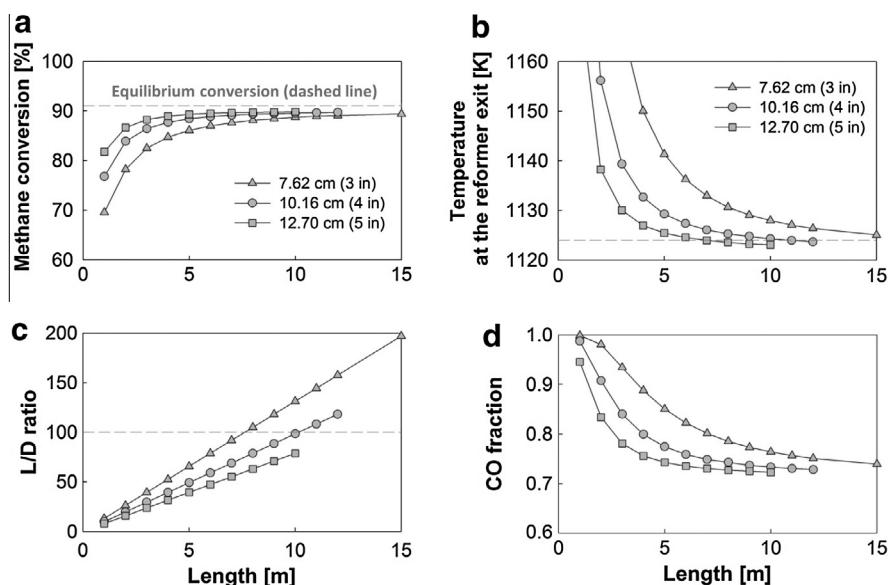


Fig. 4. Effects of tube diameter and length on (a) methane conversion, (b) temperature at the reformer exit (dashed line = threshold value of 1124 K), (c) L to D ratio (dashed line = threshold value of 100), and (d) CO fraction ($=CO/(CO + CO_2)$). Operating conditions: number of tubes = 3, pressure = 1.0 MPa, feed temperature = 1123 K, heating rate = 7.0 GJ/h, feed flow rate = 125 kmol/h (2364.82 kg/h), and a molar feed composition of $CH_4/H_2O/CO_2 = 28/66/6$.

be much larger than that of dispersion; therefore, the diameter of the tube was specified to be larger than that of the MeOH synthesis reactor (0.04 m). The simulations were conducted for a plug flow reactor with reaction rates obtained in our previous work [17] for various tube diameters and lengths by using the “Plug Flow Reactor” unit module in the UniSim Design Suite (Honeywell

Inc.), while the number of tubes was fixed to be three. As shown in Fig. 4, when the reactor length was short, the space velocity was relatively high, resulting in the conversion far from the equilibrium (Fig. 4a). Additionally, because the supplied heat was partially used for the reaction, the remaining heat increased the amount of reactor effluent (Fig. 4b), indicating that the tube

Table 2
Reactor specification, reference operating conditions, and stream information.

		Reformer		MeOH reactor		
Reactor sizing & operating conditions	Diameter [cm]	10.2		4		
	Length [m]	10		7		
	Number of tubes	3		60		
	Heat supplied [GJ/h]	7				
	Wall temperature [K]			483.2		
Local conversion [%]	CH ₄	89.55				
	CO ₂	−46.16		11.63		
	CO			39.56		
Stream name ^a						
	Feed	Reformer effluent	MeOH inlet	MeOH outlet	Separator liquid out	
Molar flow [kmol/h]	H ₂ O	83	48.4	0.46	1.68	1.68
	CH ₄	35	3.66	3.66	3.66	1.41E−02
	CO ₂	7.0	10.2	10.2	9.03	0.36
	CO		28.1	28.1	17.0	0.02
	H ₂		97.3	97.3	71.5	4.08E−02
	MeOH				12.2	11.9
	DME				0.03	5.90E−03
	Sum	125.00	187.7	139.7	115.1	14.0
Mass flow [kg/h]	H ₂ O	1495	872.4	8.32	30.3	30.2
	CH ₄	561.5	58.7	58.7	58.7	0.23
	CO ₂	308.1	450.3	449.6	397.3	15.8
	CO		787.5	787.5	476.0	0.44
	H ₂		196.1	196.1	144.1	8.22E−02
	MeOH				392.3	380.7 ^b
	DME				1.47	0.27
	Sum	2365	2365	1500	1500	428
	Temperature [K]	1123	1124	519	506	298
	Pressure [MPa]	1	5	5	5	5

^a Red-colored streams in Fig. 3b.

^b 380.7 kg-MeOH/h = 9.14 TPD.

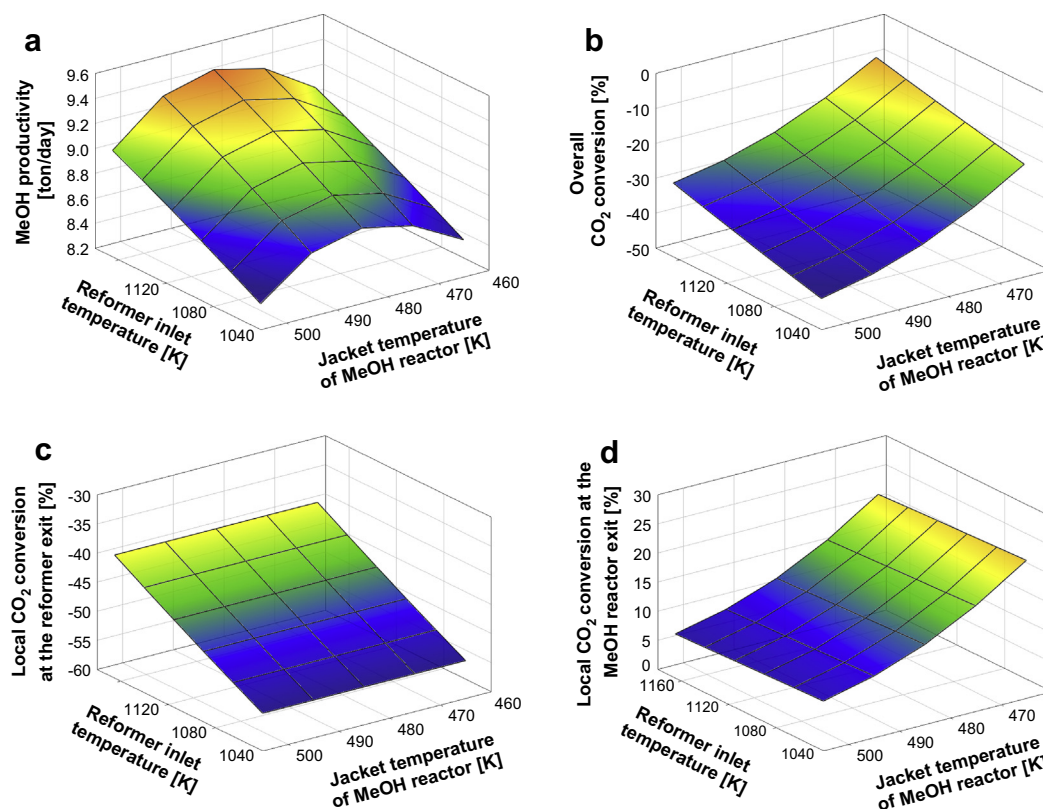


Fig. 5. Effects of reformer inlet temperature and jacket temperature of MeOH reactor on (a) MeOH productivity, (b) overall CO₂ conversion [%], defined as $((CO_{2,feed} - CO_{2,MeOH\ exit})/CO_{2,feed}) \times 100$, (c) local CO₂ conversion at the reformer exit [%], defined as $((CO_{2,feed} - CO_{2,reformer\ exit})/CO_{2,feed}) \times 100$, and (d) local CO₂ conversion at the MeOH reactor exit [%], defined as $((CO_{2,MeOH\ in} - CO_{2,MeOH\ exit})/CO_{2,MeOH\ in}) \times 100$. Operating conditions: pressure = 1.0 MPa, and a molar feed composition of CH₄/H₂O/CO₂ = 28/66/6.

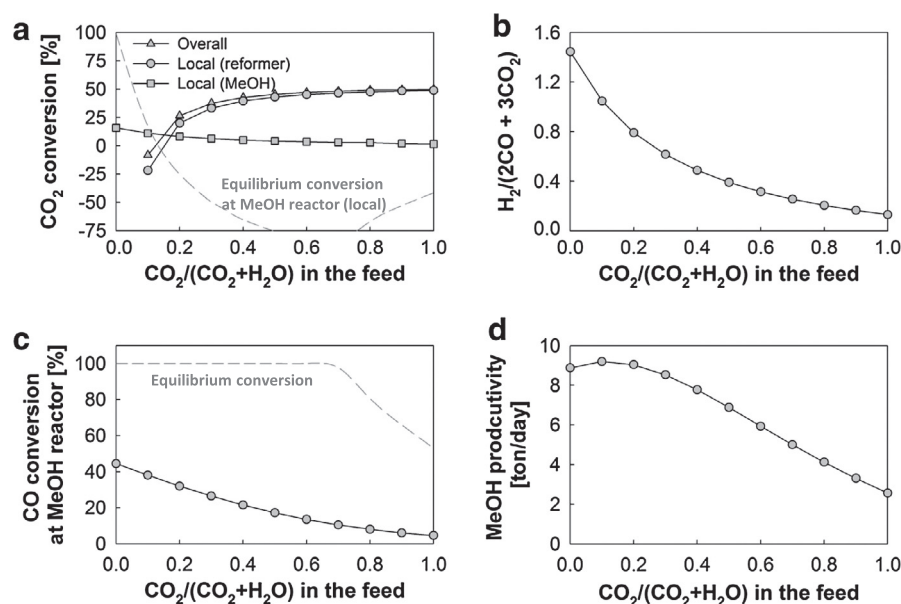


Fig. 6. Effects of CO₂ fraction ($=CO_2/(CO_2 + H_2O)$) in the feed on (a) CO₂ conversion [%], (b) H₂ fraction ($=H_2/(2CO + 3CO_2)$) at the reformer exit, (c) CO conversion at MeOH reactor [%], and (d) MeOH productivity. Operating conditions: pressure = 1.0 MPa, reformer inlet temperature = 1123 K, and jacket temperature = 483 K in the MeOH reactor. Feed flow rates of methane and (H₂O + CO₂) are 35 and 90 kmol/h, respectively.

diameter needs to be bigger than 10.16 cm (4 in). If the length-to-diameter ratio (L/D) is assumed to be close to 100, which is a usual value in industrial reformers for the ammonia synthesis process [35], the diameter of the tube is 10.16 cm, and its length is 10 m

(L/D ratio = 98.4) in this work. As shown in Fig. 4d, the CO fraction was higher than 0.7 for all the specifications; thus, the condition CO fraction = 0.5 could not be satisfied. Finally, the respective sizes and operating conditions of the reformer and the MeOH synthesis

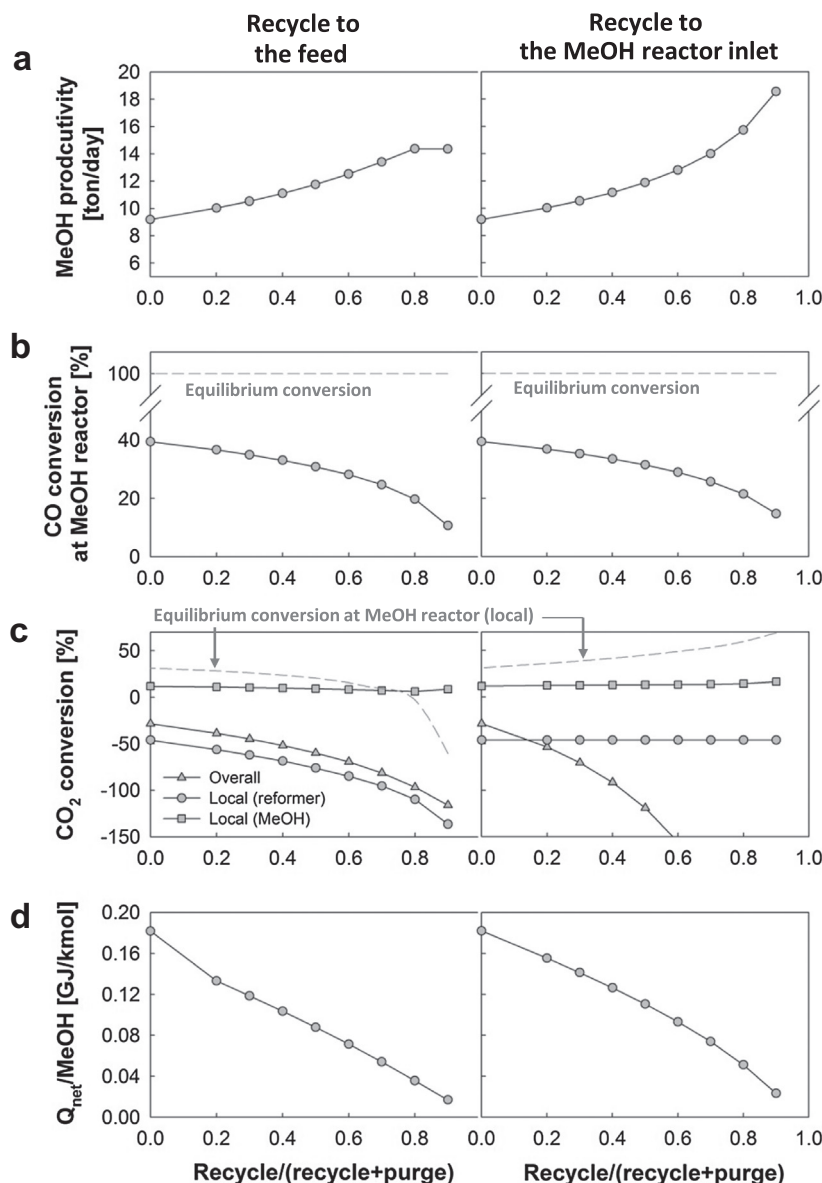


Fig. 7. Effects of recycle fraction (=recycle/(recycle + purge)) on (a) MeOH productivity, (b) CO conversion at MeOH reactor [%], (c) CO₂ conversion [%], and (d) the net heat supplied to the plant divided by the MeOH productivity [GJ/kmol] for the two separate cases of recycle to the feed (left column) and recycle to the MeOH reactor inlet (right column).

reactor were applied to the process simulator without recycle (Fig. 3b), and the resulting stream information is provided in Table 2.

3.3. Effects of operating conditions

The change of the MeOH productivity with respect to the reforming and the MeOH synthesis temperatures is shown in Fig. 5a, where the MeOH productivity increased with increasing reforming temperature because a high reforming temperature caused a high production rate of syngas for both kinetic and thermodynamic reasons. In the case of the MeOH synthesis, when the temperature was low, the reaction was dependent on the reaction rate because it was far from the equilibrium (in other words, the MeOH production rate in the kinetic-dependent domain increased with the increase of temperature). Meanwhile, when the temperature increased further, the equilibrium began to affect the conversion (in other words, the reaction now occurred in a

thermodynamic-dependent domain), and the reaction rate decreased with increasing temperature because of the exothermicity of the reaction. As for the CO₂ consumption, the increase of the reforming temperature resulted in an increase in the overall CO₂ conversion (Fig. 5b), defined as $((\text{CO}_{2,\text{feed}} - \text{CO}_{2,\text{MeOH exit}})/\text{CO}_{2,\text{feed}}) \times 100$, mostly due to the reduced production of CO₂ in the reformer (Fig. 5c). The CO₂ conversion in the MeOH reactor was unaffected by the change in the reforming temperature (Fig. 5d), although it was dependent on the jacket temperature of the MeOH reactor. Accordingly, when the reforming temperature and the MeOH synthesis temperature were maintained to be high and low, respectively, the amount of CO₂ produced (negative overall conversion) showed the minimum value (Fig. 5b). However, such an operating condition could not guarantee the highest MeOH production rate (Fig. 5a), indicating that different operating conditions should be considered to achieve an optimum balance between achieving the maximum MeOH production and the minimum CO₂ production.

The CO₂ fraction, defined as CO₂/(CO₂ + H₂O), in the feed was varied since it significantly influences the amount of CO₂ consumption. As shown in Fig. 6a, the increase of CO₂ fraction in the feed enhanced the rate of dry (CO₂) reforming and switched CO₂ production to CO₂ consumption. A further increase of the CO₂ fraction resulted in the increase of both the overall and local CO₂ conversions in the reformer. However, because a relatively higher fraction of dry reforming occurred compared with the fraction of steam reforming, there was a decrease in the H₂ fraction (the H₂/CO ratios for the dry and steam reforming processes were 1 and 3, respectively). As shown in Fig. 6b, both the CO and CO₂ conversions in the MeOH reactor decreased with increasing CO₂ fraction, and the maximum MeOH production rate was achieved for a low CO₂ fraction in the feed (Fig. 6d). This result also shows that, as in our previous work on the mixed reforming [17], there exists a trade-off between the maximum MeOH synthesis and maximum CO₂ conversion.

Equilibrium conversions for CO and CO₂ were provided in Fig. 6a and c. Due to high fraction of H₂, the equilibrium CO conversion was close to 100% when CO₂ fraction is low, and then it decreased for CO₂ fraction higher than 0.6. Since the flow rate was high enough, CO conversion at the MeOH reactor outlet was lower than the equilibrium conversion for entire range of CO₂ fraction, indicating that the conversion was not limited by the equilibrium. Meanwhile, the equilibrium CO₂ conversion shows complicated behavior since there exist reactions for both CO₂ consumption (CO₂ hydrogenation) and generation (WGS reaction) (see Fig. 3a for overall reactions). When CO₂ fraction in the feed (at the inlet of the reformer) is low, the equilibrium CO₂ conversion is positive because CO₂ hydrogenation is more dominant than WGS reaction at equilibrium, while the opposite behavior is observed for high CO₂ fraction; thus, CO₂ conversion should be lower than the equilibrium conversion when it is positive (no more CO₂ is consumed than the equilibrium), whereas the conversion should be higher than negative equilibrium conversion (no more CO₂ is produced than the equilibrium). When CO₂ fraction in the feed is close to 0.1, the CO₂ conversion reaches the equilibrium (limited by the equilibrium), but, the other values maintained the conversion far from the equilibrium.

3.4. Effect of recycle

Two cases were considered for the recycle of unreacted reactants: a recycle to the feed inlet (red colored line in Fig. 3c) or a recycle to the MeOH reactor inlet (green colored line in Fig. 3c). For the recycle to the feed inlet, the increase in the fraction of the recycle stream, defined as recycle/(recycle + purge), increased the MeOH productivity (left; Fig. 7a). It should be noticed that, a recycle to the feed inlet resembles existing industrial plants except that CO₂ is injected at the inlet of the reformer. This type of operation has several benefits in the sense that it enables to avoid adjusting the CO₂ content at the upstream of the methanol loop. In addition to this layout which is widely used in the industry, a recycle to the MeOH reactor was considered since different layout may result in different CO₂ utilization. Another benefit of the recycle is that, since the methanol production is limited by the chemical equilibrium, the use of a recycle flow can prevent the related problems, particularly at high temperature. Although both CO and CO₂ conversions were not limited by the equilibrium in the MeOH reactor without recycle stream in the present study (for details, refer to Fig. 6 and the corresponding discussion), recycle was considered to see its effect on the CO₂ utilization.

Obviously, the recycle stream increased the amount of total feed, and consequently the CO conversion was decreased because of the decreased residence time. However, since productivity is the product of feed flow rate and conversion, the net productivity

was increased at the expense of the separation cost. Additionally, the increase in the amount of the recycled reactants increased the amount of CO₂ produced in the reformer. Meanwhile, the net heat flow supplied to the process (the sum of all the heat flows) per mole MeOH produced significantly decreased with increasing recycle ratio; this relationship indicates that CO₂ emission can be decreased in an indirect manner by reducing the consumption of fossil fuels for heat supply. Overall, the benefits of recycling are the increase in the MeOH productivity and the decrease of CO₂ emission by fossil fuels, while the disadvantages are the respective increases in CO₂ generation in the reforming reaction and separation cost of unreacted reactants (related to the increase of CO₂ emission due to the increased energy usage in the separator).

In the case of the recycle to the MeOH reactor inlet, a similar dependence on the recycle ratio was observed, compared to the recycle to the feed inlet. However, although an increased recycle ratio increased the MeOH productivity to a higher value than that observed for the recycle to the feed (right, Fig. 7a), the CO₂ production significantly increased (right, Fig. 7c). Therefore, a balance existed between the maximum MeOH production and minimum CO₂ production, based on the location and amount of recycle selected.

4. Conclusions

The effect of changes in the mixed reforming rate in our previous work was corroborated using experimental data in a bench-scale reformer, and a methanol synthesis kinetic model was developed for use in a large-scale reactor with either a powder- or pellet-type catalyst. The resulting reforming kinetic model was useful in the design of a large-scale reformer, and a MeOH synthesis process, which comprised two reactions to achieve the conversion of methane to MeOH, was quantitatively analyzed with the help of kinetic models. In addition to the effects of operating conditions on MeOH production rate, an analysis of CO₂ utilization was performed, and it was clearly shown that a balance could be achieved between the maximum MeOH production and maximum CO₂ utilization (minimum CO₂ production).

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Appendix A. Supplementary materials

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.fuel.2014.03.068>.

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